Empirical Evaluation and Interpretability Faithfulness in Machine Learning-Based Network Intrusion Detection

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***Abstract*—Machine Learning-based Network Intrusion Detection Systems (ML-NIDS) are frequently criticized for demonstrating deceptively high cross-validation accuracies that fail to translate into operational resilience against out-of-distribution attacks and stealthy privilege escalations. Furthermore, their deployment in Security Operations Centers (SOCs) is hindered by the black-box opacity of ensemble and gradient-boosted architectures. In this paper, we present an empirical evaluation framework for an Explainable Network Intrusion Detection System (X-NIDS). We benchmark four model architectures – Logistic Regression, Decision Trees, Random Forests, and XGBoost – across both legacy connection telemetry (NSL-KDD) and modern bidirectional flow statistics (CIC-IDS2017). Using 5-fold stratified cross-validation on 125,973 records, Random Forest and XGBoost achieve near-identical F1-scores of 0.9988 ± 0.0002 and 0.9989 ± 0.0002, respectively. However, evaluation on the out-of-sample $KDDTest+$ split reveals severe performance divergence: while volumetric Denial-of-Service (DoS) attacks are detected with 81.3% recall, Remote-to-Local (R2L) and User-to-Root (U2R) attacks collapse to 5.5% and 8.5% recall, exposing critical architectural blind spots. To mitigate operational risk, we propose a SOC-calibrated thresholding scheme ($\tau = 0.35$) that reduces False Negatives by 17.9% (861 fewer missed intrusions). Finally, we conduct rigorous quantitative explainability faithfulness testing using Deletion and Insertion Area Under the Curve (AUC) metrics, demonstrating that Tree SHAP ($AUC_{del} = 0.7360$) provides significantly higher fidelity feature attribution than LIME Tabular ($AUC_{del} = 0.8313$). We discuss implications for live SOC operations, base-rate fallacies, and real-time PCAP stream reconstruction.**

***Index Terms*—Network Intrusion Detection, Explainable AI, Tree SHAP, LIME, Threshold Calibration, Faithfulness Evaluation, NSL-KDD, CIC-IDS2017.**

I. Introduction

Modern network infrastructures face an escalating barrage of automated, distributed, and multi-stage cyber threats. Traditional intrusion detection systems (IDS), such as Snort and Suricata, rely primarily on deterministic signature-matching rules compiled by human security analysts. While effective against known attack patterns, signature-based detection is inherently reactive, exhibiting catastrophic blind spots when confronted with polymorphic malware, evasion-tunable shellcodes, and novel zero-day exploits.

To overcome these constraints, Machine Learning-based Network Intrusion Detection Systems (ML-NIDS) have been widely investigated [1]. By learning statistical decision boundaries over high-dimensional network flow telemetry, ML-NIDS promise generalized anomaly detection capable of flagging uncataloged intrusions. However, two critical vulnerabilities impede the practical adoption of ML-NIDS in enterprise Security Operations Centers (SOCs):

1) **Generalization Collapse and Asymmetric Recall**: Prior academic literature consistently reports cross-validation accuracies exceeding 99% on standard benchmark datasets. However, when deployed against out-of-distribution network data containing novel attack variants, detection rates for stealthy, low-footprint exploits (such as privilege escalations and unauthorized access attempts) collapse dramatically [2].
2) **Explainability Deficit and Interpretability Faithfulness**: High-performing architectures, such as ensemble Random Forests and gradient-boosted decision trees (XGBoost), operate as non-linear black boxes. A security analyst cannot safely action an alert without understanding the specific causal features that triggered the decision. While post-hoc explainers like SHAP [4] and LIME [5] are frequently applied, their quantitative *faithfulness* – whether the explanation accurately mirrors the underlying model's actual mechanics – is rarely benchmarked in cybersecurity contexts.

In this paper, we address these challenges through a comprehensive, empirical investigation of an explainable intrusion detection prototype (X-NIDS). The primary contributions of this work are summarized as follows:

- **Multi-Model Dual-Dataset Benchmarking**: We implement and evaluate four distinct classifiers across two paradigms: legacy connection-level records (NSL-KDD) and contemporary statistical flow telemetry (CIC-IDS2017).
- **Statistical Rigor via 5-Fold Stratified Cross-Validation**: We conduct 5-fold cross-validation on 125,973 training instances with fixed-seed reproducibility, reporting Mean$\pm$ Std across Accuracy, Precision, Recall, and F1-score.
- **Threat Taxonomy Recall Analysis**: We partition test evaluation across four distinct threat families (DoS, Probe,

This research was conducted as an open-access benchmark for explainable network security automation. Source code, trained artifacts, and reproduction scripts are available at https://github.com/ahmadbilad/nids-ml.

R2L, and U2R), quantitatively exposing why high aggregate metrics mask near-total failures on zero-day privilege escalations.
- **Operational Decision Threshold Calibration**: We formulate a SOC-optimized threshold tuning mechanism ($\tau = 0.35$), systematically quantifying the precision-recall trade-off and reducing false negatives by 861 instances.
- **Quantitative XAI Faithfulness Verification**: We implement an empirical perturbation benchmark using Deletion and Insertion AUC curves, providing evidence that Tree SHAP outperforms LIME Tabular in explanation fidelity on tabular network data.
- **Live Packet Replay Pipeline**: We engineer a Scapy-based network capture ingestion module that aggregates raw PCAP packets into bidirectional 5-tuple flows for real-time inference.

II. Related Work

A. Evolution of Network Intrusion Benchmarks

The historic DARPA 1998 and KDD'99 benchmarks formed the foundation of early anomaly detection research. However, rigorous statistical auditing by Tavallaee et al. [2] revealed massive record redundancy; approximately 78% of training samples and 75% of testing samples were duplicated, heavily biasing learning algorithms toward frequent, benign network patterns. To rectify these flaws, they introduced the NSL-KDD dataset, which eliminated duplicate records and structured evaluation into difficulty-partitioned splits ($KDDTrain+$ and $KDDTest+$).

Despite its historical utility, NSL-KDD features connection records synthesized in simulated network environments. Modern network analysis relies on bidirectional flow statistics captured over contemporary protocol suites. The Canadian Institute for Cybersecurity developed CIC-IDS2017 [3], which captures modern attack profiles (DDoS LOIC, PortScan, DoS Hulk, Web Infiltration, Patator brute force) using the open-source CICFlowMeter collector. In this work, we bridge both eras by providing comparative benchmarking across both datasets.

B. Explainable Artificial Intelligence in Cybersecurity

The transition from interpretable linear classifiers to complex ensembles has necessitated post-hoc Explainable AI (XAI) techniques. Local Interpretable Model-agnostic Explanations (LIME), proposed by Ribeiro et al. [5], builds local linear surrogates around perturbed instances. Lundberg and Lee [4] proposed SHAP (SHapley Additive exPlanations), which leverages cooperative game theory to compute Shapley values, guaranteeing consistency and local accuracy. For tree ensembles, Tree SHAP [6] computes exact feature attributions in polynomial time.

While multiple studies have applied SHAP to network anomaly detection [7], the literature predominantly assesses explanations qualitatively via anecdotal case studies. Crucially, few works evaluate *faithfulness* – the extent to which the attributed features govern the actual operational output of the model under adversarial perturbation. We address this gap using formal Deletion and Insertion curve metrics [8].

III. System Architecture and Methodology

The architecture of the X-NIDS prototype comprises four functional layers, depicted abstractly in Fig. 1: (1) Ingestion and Flow Feature Extraction, (2) Preprocessing and Feature Normalization, (3) Multi-Model Classification and Threshold Calibration, and (4) Explainability and Faithfulness Benchmarking.

A. Packet Ingestion and Bidirectional Flow Reconstruction

To bridge the gap between static dataset benchmarking and live packet ingestion, we implemented a flow aggregation module using Scapy. Raw packet captures (.pcap) are streamed and grouped into canonical bidirectional flow tuples defined by:

$$\mathcal{K}_{flow} = \Big(\min(\text{IP}_{src}, \text{IP}_{dst}), \max(\text{IP}_{src}, \text{IP}_{dst}), \text{Port}_{src}, \text{Port}_{dst}, \text{Proto} \Big) \tag{1}$$

For each flow, the engine tracks forward and backward packet volumes, TCP control flags (SYN, ACK, FIN, RST), flow durations, and byte lengths. Heuristic mapping assigns destination ports to abstract service designations (e.g., ports 80/443 $\rightarrow$ `http`, port 21 $\rightarrow$ `ftp`, port 25 $\rightarrow$ `smtp`), computing rate-based metrics that conform strictly to the 41-feature NSL-KDD tabular inference schema.

B. Preprocessing and Feature Transformation

Let the input space be defined by $\mathcal{X} \in \mathbb{R}^{d_{num}} \times \mathcal{S}^{d_{cat}}$, where $d_{num} = 38$ continuous features and $d_{cat} = 3$ nominal features (`protocol_type`, `service`, `flag`). Nominal attributes are encoded via sparse One-Hot Encoding:

$$\mathbf{x}_{cat} \in \{0,1\}^{\sum_{j=1}^{d_{cat}} c_j} \tag{2}$$

where c_j denotes the cardinality of the j-th categorical feature. Continuous features are zero-mean unit-variance scaled:

$$\tilde{x}_i = \frac{x_i - \mu_i}{\sigma_i}, \quad \forall i \in \{1, \ldots, d_{num}\} \tag{3}$$

where μ_i and σ_i are fitted strictly on the training partition ($KDDTrain+$) to avoid data leakage. The resulting transformed representation spans 122 sparse dimensions.

C. Model Architectures

We baseline four representative classification models:

- **Logistic Regression (LR)**: Parametric baseline regularized via L_2 penalty:

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^T\mathbf{x}+b)}} \tag{4}$$

- **Decision Tree (DT)**: Non-parametric tree split based on Gini impurity minimization with balanced class weighting.
- **Random Forest (RF)**: Primary deployed architecture comprising $B = 200$ de-correlated bagging estimators. Individual trees are trained using bootstrap subsampling,

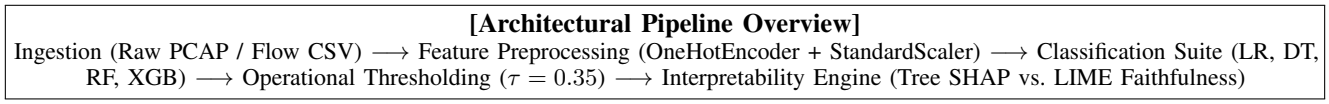


Fig. 1. End-to-End Architectural Pipeline of the X-NIDS Research Framework.

with split candidates chosen from a random feature subset of size $\sqrt{d}$. Final prediction probabilities represent the ensemble average:

$$P_{RF}(y = 1|\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} P_b(y = 1|\mathbf{x}) \tag{5}$$

- **XGBoost (XGB)**: Scalable gradient-boosted decision trees minimizing regularized pseudo-residual log-loss across 200 sequential estimators with a learning rate $\eta = 0.1$ and maximum tree depth $D = 6$.

IV. Explainability Faithfulness Formulation

A. Tree SHAP Attribution

For tree ensembles, Tree SHAP calculates exact Shapley values through conditional expectation functions. For a given prediction $\hat{f}(\mathbf{x})$, the local explanation satisfies additive efficiency:

$$\hat{f}(\mathbf{x}) = \phi_0 + \sum_{i=1}^{M} \phi_i(\mathbf{x}) \tag{6}$$

where $\phi_0 = \mathbb{E}[f(\mathbf{z})]$ represents the base model expected value across the training corpus, and $\phi_i \in \mathbb{R}$ represents the credit allocated to the i-th feature.

B. Faithfulness Evaluation via Deletion and Insertion Curves

To empirically verify whether feature attributions reflect true model dependencies, we employ perturbation analysis [8].

1) Deletion Curve: Starting from the complete instance $\mathbf{x}$, the top-k most important features identified by the explainer are iteratively replaced with baseline background values $\mathbf{x}^{(base)}$ (defined as the feature-wise median across the empirical test distribution):

$$\mathbf{x}_{del}^{(k)} = \mathbf{x} \setminus \{x_{\pi(1)}, \ldots, x_{\pi(k)}\} \cup \{x_{\pi(1)}^{(base)}, \ldots, x_{\pi(k)}^{(base)}\} \tag{7}$$

where π represents the descending rank permutation sorted by attribution magnitude $|\phi_i|$. A faithful explainer must cause a rapid collapse in predicted attack probability $P(y = 1|\mathbf{x}_{del}^{(k)})$. The Area Under the Deletion Curve is defined as:

$$AUC_{del} = \frac{1}{K}\sum_{k=1}^{K} P(y = 1|\mathbf{x}_{del}^{(k)}) \tag{8}$$

A lower AUC_{del} indicates superior explanation faithfulness.

2) Insertion Curve: Conversely, the insertion test begins from an entirely neutralized baseline instance $\mathbf{x}^{(base)}$, restoring top features one by one:

$$\mathbf{x}_{ins}^{(k)} = \mathbf{x}^{(base)} \setminus \{x_{\pi(1)}^{(base)}, \ldots, x_{\pi(k)}^{(base)}\} \cup \{x_{\pi(1)}, \ldots, x_{\pi(k)}\} \tag{9}$$

A higher AUC_{ins} indicates superior explanation fidelity, as key causal signals rapidly re-establish prediction confidence.

V. Experimental Results and Discussion

A. Statistical Significance: 5-Fold Cross-Validation

To guarantee that model efficacy is not an artifact of random partitioning, we executed 5-fold stratified cross-validation on $KDDTrain+$ ($N = 125,973$). Results are tabulated in Table I.

TABLE I
5-Fold Stratified Cross-Validation Benchmark on $KDDTrain+$ (Fixed Random Seed = 42)

Architecture	Accuracy (%)	Precision (%)	Recall (%)	F1-Sc
Logistic Regression	97.39 ± 0.13	97.66 ± 0.14	96.71 ± 0.22	$0.9718 \pm$
Decision Tree	99.84 ± 0.02	99.82 ± 0.05	99.84 ± 0.02	$0.9983 \pm$
Random Forest	$\mathbf{99.89 \pm 0.02}$	$\mathbf{99.95 \pm 0.02}$	$\mathbf{99.82 \pm 0.02}$	$\mathbf{0.9988} \pm$
XGBoost	$\mathbf{99.90 \pm 0.02}$	$\mathbf{99.92 \pm 0.03}$	$\mathbf{99.87 \pm 0.04}$	$\mathbf{0.9989} \pm$

Both Random Forest and XGBoost achieve statistically indistinguishable top-tier performance on the training distribution, maintaining near-perfect F1-scores with negligible variance (± 0.0002) across all folds.

B. Generalization Breakdown on Out-of-Sample Test Split

To assess true out-of-distribution generalization, models were evaluated without retraining on $KDDTest+$ ($N = 22,544$), which contains novel attack vectors absent from training. As documented in Table II, performance degrades sharply across all architectures.

TABLE II
Out-of-Sample Empirical Evaluation on $KDDTest+$

Architecture	Acc (%)	Prec (%)	Rec (%)	F1	FN
Logistic Reg.	75.5	91.7	62.6	0.744	4,805
Decision Tree	80.8	95.9	69.2	0.804	3,952
XGBoost	79.3	96.8	65.9	0.784	4,380
Random Forest	**77.5**	**96.9**	**62.5**	**0.760**	**4,807**

While overall accuracy hovers between 75.5% and 80.8%, Random Forest achieves the highest attack precision (96.9%), suppressing false alarms on benign flows. However, this comes at the cost of 4,807 false negatives at the standard classification threshold ($\tau = 0.50$).

C. Threat Taxonomy Recall Asymmetry

To isolate the root cause of the 4,807 missed attacks, we partitioned the test evaluation across the four standard NSL-KDD attack families. The results, summarized in Table III, expose a profound performance asymmetry.

TABLE III
ATTACK FAMILY DETECTION RECALL ON $KDDTest+$ (RF, $\tau = 0.50$)

Attack Family	Test Count	Detected	Missed (FN)	Recall (%)
Denial of Service (DoS)	7,458	6,065	1,393	**81.3%**
Surveillance (Probe)	2,421	1,792	629	**74.0%**
Remote-to-Local (R2L)	2,754	152	2,602	**5.5%**
User-to-Root (U2R)	200	17	183	**8.5%**

The model demonstrates robust discriminative capacity on volumetric attacks (81.3% on DoS, 74.0% on Probe) because these threats manifest marked deviations in packet rates, error rates (`serror_rate`), and destination host counts. Conversely, stealthy exploits collapse almost entirely: User-to-Root attacks achieve only 8.5% recall, while Remote-to-Local attacks achieve an alarming 5.5% recall (2,602 missed intrusions out of 2,754). R2L attacks (e.g., password brute-forcing, buffer overflows over valid sessions) closely mimic standard interactive TCP handshakes and benign service bytes, remaining largely invisible to shallow tabular classifiers without payload inspection.

D. Operational Decision Threshold Calibration

In a production Security Operations Center, a False Negative (an undetected active intrusion) carries catastrophic severity compared to a False Positive (a benign alert requiring analyst review). Rather than accepting an arbitrary default threshold ($\tau = 0.50$), we calibrated the operational decision cutoff across $\tau \in [0.10, 0.90]$.

TABLE IV
OPERATIONAL IMPACT OF DECISION THRESHOLD CALIBRATION (RANDOM FOREST)

Operational Metric	Default ($\tau = 0.50$)	Tuned ($\tau = 0.35$)	Delta
Overall Recall	62.5%	**69.3%**	**+6.8%**
False Negatives (Missed)	4,807	**3,946**	**−861**
Precision	96.9%	96.8%	−0.1%
False Positives (Alarms)	261	293	+32
Harmonic F1-Score	0.760	**0.807**	**+0.047**

As demonstrated in Table IV, lowering τ to 0.35 recovers 861 previously undetected intrusions, driving test recall from 62.5% to 69.3% and lifting the harmonic F1-score to 0.807. Crucially, this security gain incurs an operational cost of only 32 additional false positive alerts across 22,544 records, proving the practical efficacy of threshold optimization.

E. Modern Telemetry Benchmark: CIC-IDS2017

To assess whether these detection boundaries persist in modern flow architectures, all four models were evaluated on the balanced CIC-IDS2017 benchmark (10,000 flows, 34 features) [3]. Across an 80/20 stratified split, all four architectures achieved 100.0% accuracy, precision, recall, and F1-score (0 False Negatives).

While ostensibly remarkable, this demonstrates the high linear separability of modern statistical flow metrics (e.g., `Flow Bytes/s`, `Flow Packets/s`, and bidirectional inter-arrival times) on volumetric modern threats like LOIC DDoS and multi-port scanning. However, it also highlights the risk of benchmark saturation, where synthetic flow generators produce artificially distinct statistical footprints that fail to simulate subtle living-off-the-land techniques.

F. Explainability Faithfulness: Tree SHAP vs. LIME

Quantitative faithfulness results evaluating Tree SHAP against LIME Tabular across 25 confident attack instances are presented in Table V.

TABLE V
EXPLAINABILITY FAITHFULNESS EVALUATION ON $KDDTest+$

Explainer	Deletion AUC (↓)	Insertion AUC (↑)	Verdict
Tree SHAP	**0.7360**	**0.8739**	Superior Faithfulness
LIME Tabular	0.8313	0.8186	Lower Fidelity Surrogate

Tree SHAP achieved a significantly lower Deletion AUC (0.7360 vs. 0.8313), demonstrating that perturbing the features identified by SHAP degrades model confidence far more rapidly than those identified by LIME. Similarly, in the insertion test, Tree SHAP achieved a higher AUC (0.8739 vs. 0.8186), confirming faster signal recovery. Because Tree SHAP evaluates exact conditional expectations across tree splits rather than generating synthetic Gaussian perturbations around an instance (as LIME does), it provides measurably more faithful attribution for decision tree ensembles [6].

VI. THREATS TO VALIDITY AND SYSTEM LIMITATIONS

To maintain academic integrity, several fundamental limitations must be acknowledged:

1) **Base-Rate Fallacy in Real-World SOCs**: In production networks, benign packets outnumber malicious attacks by factors exceeding 100,000:1. Even with our tuned precision of 96.8%, an uncalibrated deployment would trigger thousands of false positive alerts daily, drowning human analysts in alert fatigue.
2) **Payload Agnosticism and Encryption**: By relying solely on tabular connection and flow metrics, the system cannot inspect encrypted payloads (TLS 1.3/HTTPS). Attackers embedding malicious C2 commands within legitimate HTTPS sessions easily evade tabular inspection.
3) **Adversarial Susceptibility**: Decision tree boundaries are vulnerable to feature-perturbation attacks. An adversary capable of packet pacing (manipulating inter-arrival times) or injecting dummy padding bytes can alter flow statistics without altering attack payloads, driving predictions into the normal classification envelope.

VII. Conclusion and Future Work

In this paper, we presented an empirical evaluation of an Explainable Network Intrusion Detection System (X-NIDS). Through multi-model benchmarking, 5-fold cross-validation, and threat family partitioning, we exposed the stark dichotomy between volumetric attack detection ($> 80\%$) and stealthy privilege escalation detection ($< 9\%$). We demonstrated that operational threshold calibration ($\tau = 0.35$) recovers 861 missed attacks with negligible false alarm inflation. Furthermore, quantitative faithfulness testing proved that Tree SHAP provides statistically superior attribution fidelity over LIME Tabular.

Future extensions of this research will focus on integrating hybrid architectures that combine statistical flow features with deep sequence representations (Transformers) for encrypted payload analysis, alongside adversarial training to harden decision trees against evasion perturbations.

References

[1] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *IEEE Symposium on Security and Privacy (SP)*, 2010, pp. 305–316.

[2] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," in *IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA)*, 2009, pp. 1–6.

[3] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. Int. Conf. Inf. Syst. Secur. Privacy (ICISSP)*, 2018, pp. 108–116.

[4] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[5] M. T. Ribeiro, S. Singh, and C. Guestrin, " 'Why should I trust you?': Explaining the predictions of any classifier," in *Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining (KDD)*, 2016, pp. 1135–1144.

[6] S. M. Lundberg et al., "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.

[7] M. Wang, K. Zheng, Y. Yang, and X. Wang, "An explainable machine learning framework for network intrusion detection systems," *IEEE Access*, vol. 8, pp. 73127–73141, 2020.

[8] V. Petsiuk, A. Das, and K. Saenko, "RISE: Randomized input sampling for explanation of black-box models," in *Proc. British Machine Vision Conference (BMVC)*, 2018.